

Lab 08 –

String Revisited, Formatted Output and getting and Formatting the Date and Time

Programming Exercise:

1. Construct the strings by using the string time format function `strftime ()`
 - a) ('Thursday, July 13 2018')
 - b) ('09:40 PM Central Daylight Time on 07/13/2018')
 - c) ('I will meet you on Thu July 13 at 09:40 PM.')

Source Code:

```
1  from datetime import datetime
2
3  # Define the datetime for July 13, 2018, at 09:40 PM
4  dt = datetime(year=2018, month=7, day=13, hour=21, minute=40)
5
6  # Part a: (Thursday, July 13 2018)
7  formatted_a = dt.strftime("%A, %B %d %Y")
8
9  # Part b: ('09:40 PM on 07/13/2018') - without timezone information
10 formatted_b = dt.strftime("%I:%M %p on %m/%d/%Y")
11
12 # Manually adding "Central Daylight Time" as a static string
13 formatted_b = formatted_b.replace(__old: "on", __new: "Central Daylight Time on")
14
15 # Part c: ('I will meet you on Thu July 13 at 09:40 PM.')
16 formatted_c = dt.strftime('%I will meet you on %a %B %d at %I:%M %p.')
17
18 # Print all results
19 print(formatted_a)
20 print(formatted_b)
21 print(formatted_c)
```

Output:

```
C:\Users\hp\PycharmProjects\main\venv\Scripts\python
(Friday, July 13 2018)
('09:40 PM Central Daylight Time on 07/13/2018')
('I will meet you on Fri July 13 at 09:40 PM.')
```

```
Process finished with exit code 0
```

2. Assuming that variable `forecast` has been assigned string 'It will be a sunny day today' Write Python statements corresponding to these assignments: (a) To variable `count`, the number of occurrences of string 'day' in string `forecast`. (b) To variable `weather`, the index where substring 'sunny' starts. (c) To variable `change`, a copy of `forecast` in which every occurrence of substring 'sunny' is replaced by 'cloudy'.

Source Code:

```
1  # Given variable
2  forecast = 'It will be a sunny day today'
3
4  # (a) Count occurrences of 'day' in forecast
5  count = forecast.count('day')
6
7  # (b) Find the index where 'sunny' starts
8  weather = forecast.find('sunny')
9
10 # (c) Replace 'sunny' with 'cloudy' in a copy of forecast
11 change = forecast.replace(__old: 'sunny', __new: 'cloudy')
12
13 print(count)
14 print(weather)
15 print(change)
```

Output:

```
C:\Users\hp\PycharmProjects\main\ven'
2
13
It will be a cloudy day today

Process finished with exit code 0
|
```

3. Write function `even()` that takes a positive integer `n` as input and prints on the screen all numbers between, and including, 2 and `n` divisible by 2 or by 3, using this output format: `>>> even(17) 2, 3, 4, 6, 8, 9, 10, 12, 14, 15, 16,`

Source Code:

```
1  def even(n): 1usage
2      # Initialize an empty list to store numbers divisible by 2 or 3
3      numbers = []
4
5      # Loop from 2 up to and including n
6      for i in range(2, n + 1):
7          if i % 2 == 0 or i % 3 == 0: # Check if divisible by 2 or 3
8              numbers.append(str(i)) # Add number as string to list
9
10     # Join list into a single string with commas and print
11     print(", ".join(numbers))
12
13     n = int(input("Enter a positive integer: "))
14     even(n)
```

Output:

```
C:\Users\hp\PycharmProjects\main\venv\Sci
```

```
Enter a positive integer: 16
```

```
2, 3, 4, 6, 8, 9, 10, 12, 14, 15, 16
```

```
Process finished with exit code 0
```

4. Assume variables first, last, street, number, city, state, zipcode have already been assigned. Write a print statement that creates a mailing label: John Doe 123 Main Street AnyCity, AS 09876 assuming that: >>> first = 'John' >>> last = 'Doe' >>> street = 'Main Street' >>> number = 123 >>> city = 'AnyCity' >>> state = 'AS' >>> zipcode = '09876'

Source Code:

```
1  # Assuming the variables have been assigned as follows:
2  first = 'John'
3  last = 'Doe'
4  street = 'Main Street'
5  number = 123
6  city = 'AnyCity'
7  state = 'AS'
8  zipcode = '09876'
9
10 # Print the mailing label
11 print(f"{first} {last}")
12 print(f"{number} {street}")
13 print(f"{city}, {state} {zipcode}")
```

Output:

```
C:\Users\hp\PycharmProjects\main\ven
```

```
John Doe
```

```
123 Main Street
```

```
AnyCity, AS 09876
```

```
Process finished with exit code 0
```

5. Translate each part into a Python statement using appropriate string methods:

(a) Assign to variable message the string 'The secret of this message is that it is secret.'

(b) Assign to variable length the length of string message, using operator len().

(c) Assign to variable count the number of times the substring 'secret' appears in string message, using string method count().

(d) Assign to variable censored a copy of string message with every occurrence of substring 'secret' replaced by 'xxxxxx', using string method replace().

Source Code:

```
1 # Part (a)
2 message = 'The secret of this message is that it is secret.'
3 # Part (b)
4 length = len(message)
5 # Part (c)
6 count = message.count('secret')
7 # Part (d)
8 censored = message.replace(__old: 'secret', __new: 'xxxxxx')
9
10 print("Message:", message)
11 print("Length:", length)
12 print("Count of 'secret':", count)
13 print("Censored message:", censored)
14
```

Output:

```
C:\Users\m\p\p\venv\Scripts\python.exe C:\Users\m\p\p\venv\Scripts\python.exe
Message: The secret of this message is that it is secret.
Length: 48
Count of 'secret': 2
Censored message: The xxxxxx of this message is that it is xxxxxx.

Process finished with exit code 0
```

6. Write a function `month()` that takes a number between 1 and 12 as input and returns the three-character abbreviation of the corresponding month. Do this without using an if statement, just string operations. Hint: Use a string to store the abbreviations in order. `>>> month(1) 'Jan' >>> month(11) 'Nov'`

Source Code:

```
1  def month(n): 2 usages
2      # String of month abbreviations, each 3 characters long
3      months = "JanFebMarAprMayJunJulAugSepOctNovDec"
4
5      # Calculate the starting index for the n-th month
6      return months[(n - 1) * 3: (n - 1) * 3 + 3]
7
8  # Example usage
9  print(month(1))
10 print(month(11))
```

Output:

```
C:\Users\hp\PycharmProjects\main\venv\Scripts\python.exe "C:\Users\hp\PycharmProjec
Jan
Nov
```

```
Process finished with exit code 0
```

7. Implement function cheer() that takes as input a team name (as a string) and prints a cheer as shown: >>> cheer('uitians') How do you spell winner? I know, I know! U I T I A N S ! And that's how you spell winner! Go Uitians!

Source Code:

```
1  def cheer(team_name): 1 usage
2      # Step 1: Print the first two lines
3      print("How do you spell winner?")
4      print("I know, I know!")
5
6      # Step 2: Print the team name, each letter separated by space, and followed by "!"
7      print(" ".join(team_name.upper()) + " !")
8
9      # Step 3: Print the last two lines
10     print("And that's how you spell winner!")
11     print(f"Go {team_name.capitalize()}!")
12 cheer("uitians")
```

Output:

```
C:\Users\hp\PycharmProjects\main\venv\
How do you spell winner?
I know, I know!
U I T I A N S !
And that's how you spell winner!
Go Uitians!

Process finished with exit code 0
```